

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

PHYSICS

PAPER 2 Theory

4023/2

2 hours 15 minutes

SPECIMEN PAPER

Additional materials:
Electronic calculator
Answer booklet

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top.

At the end of the examination fasten the answer booklet used securely to the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question.

This question paper consists of two compulsory sections, **A** and **B**.

FOR EXAMINER'S USE	
Section A	
Section B	9
	10
	11
	12
TOTAL	

This specimen paper consists of 13 printed pages.

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Section A

Answer **all** questions.

- 1 (a) **Fig. 1.1** shows a measuring instrument used to measure the diameter of a glass rod.

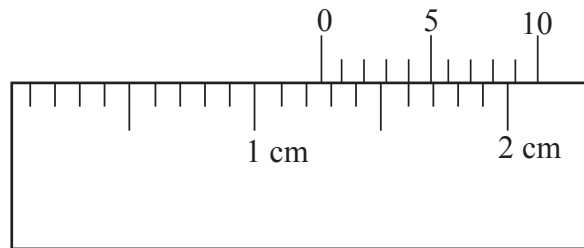


Fig. 1.1

- (i) Name the instrument in **Fig. 1.1**.

_____ [1]

- (ii) Determine the diameter of the glass rod.

_____ [2]

- (iii) Give **two** precautions taken when using the instrument.

 _____ [2]

- 2 (a) A cheetah hid in the tall grass in order to attack a grazing kudu. The cheetah accelerated uniformly at 10 m/s^2 towards the prey, 200 m away. It caught the kudu unaware with a velocity of 63 m/s.

(i) Define *acceleration*.

[1]

(ii) Determine the time taken to catch the kudu.

[2]

(iii) Describe the motion of the cheetah.

[2]

3 **(a)** Define *force*.

[1]

(b) Two fighting male goats collided head on. One goat exerted a force of 250 N while the other one exerted a force of 300 N in the opposite direction.

(i) Determine the resultant force acting in the collision.

[2]

(ii) Explain how the goat's mass could influence the actual motion observed.

[2]

4 **(a)** **(i)** State the law of conservation of *energy*.

[1]

(ii) Compare renewable and non-renewable energy sources.

[2]

- (b) Lithium mined in Zimbabwe is used to manufacture Lithium-ion batteries.

Suggest **two** safe methods that can be used in the community to dispose used lithium-ion batteries.

[2]

- 5 (a) Distinguish between a *single fixed pulley* and a *single moveable pulley*.

[2]

- (b) State the effect of increasing the length of plier handles on the

(i) velocity ratio,

[1]

(ii) mechanical advantage.

[1]

- (c) Suggest any **one** way of improving the efficiency of a machine.

[1]

- 6 (a) Give any **two** tools used in robot design.

[2]

- (b) Describe the function of an electric actuator.

[3]

- 7 (a) State the function of a

- (i) capacitor,

[1]

- (ii) reed switch,

[1]

- (iii) light emitting diode.

[1]

- (b) Fig. 7.1 represents a carbon resistor.

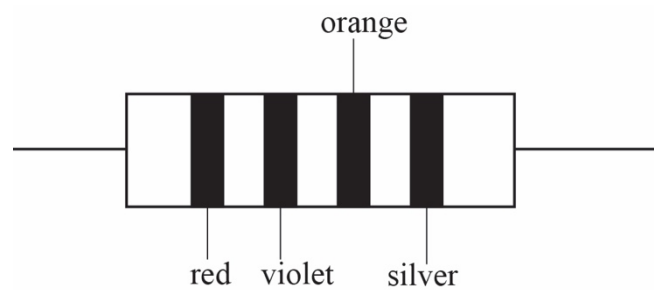


Fig. 7.1

Determine the resistance of the carbon resistor shown in Fig. 7.1.

[2]

- 8 (a) Define the term *half-life*.

[1]

- (b) Fig. 8.1 shows the decay curve of radium.

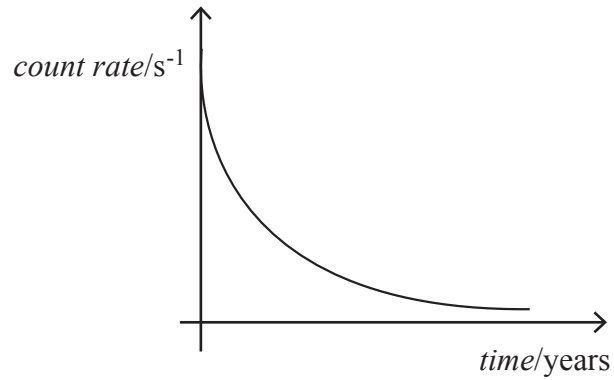


Fig. 8.1

- (i) On Fig. 8.1, draw a graph of a sample with a longer half-life than that of radium. [2]
- (ii) Name any **two** precautions that should be observed when handling radioactive materials.

[2]

Section B

Answer any **three** questions from this section.

- 9 (a) (i) State **one** use of triangles in trusses. [1]
- (ii) Distinguish between a *truss* and a *beam*. [2]
- (iii) **Fig. 9.1** shows a roof truss of a house.

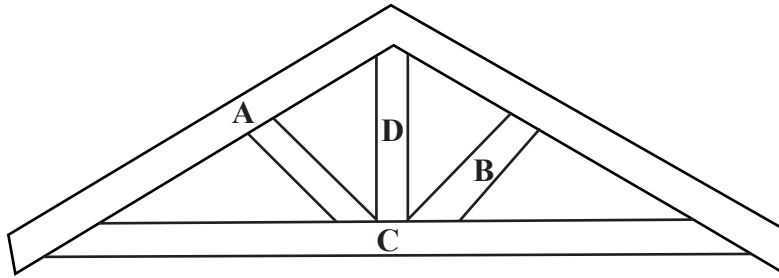


Fig. 9.1

Identify the forces acting on the beams of a roof truss in parts **A**, **B**, **C** and **D**. [4]

- (b) (i) Describe the changes that occur to volume when
1. solid water is heated, [2]
 2. water at 0°C is cooled, [2]
 3. liquid ethanol evaporates. [2]

- (ii) **Fig. 9.2** shows an experiment to determine the heat capacity of a 2 kg aluminium block.

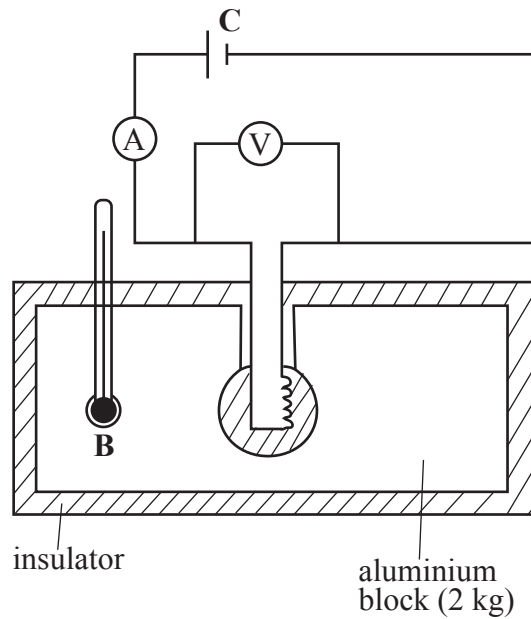


Fig. 9.2

Name the components labelled **B** and **C**. [2]

- (iii) Suggest any **two** changes to the experiment that improves the accuracy of the experiment. [2]

- (iv) The aluminium block is heated by a heater of 1 000 W for 5 minutes and the temperature changes by 20 °C.

Calculate the specific heat capacity of the aluminium block. [3]

- 10 (a) (i) Name the four strokes of a petrol engine. [4]
- (ii) Describe the operation of a compression stroke in a petrol engine. [3]
- (iii) Explain the function of a carburetor. [2]
- (b) (i) Distinguish between a *real image* and a *virtual image*. [1]
- (ii) Define the *optical centre of a converging lens*. [1]
- (iii) An object of height 2.0 cm is placed 2.5 cm away from a thin converging lens of focal length 5.0 cm.
- Draw a ray diagram to show the position of the object and the image. [3]
- (iv) Using the diagram, deduce the
1. characteristics of the image, [2]
 2. position of the image, [1]
 3. size of the image. [1]
- (v) Determine the magnification. [2]

- 11 (a) **Fig. 11.1** shows a lightning rod connected to the ground.

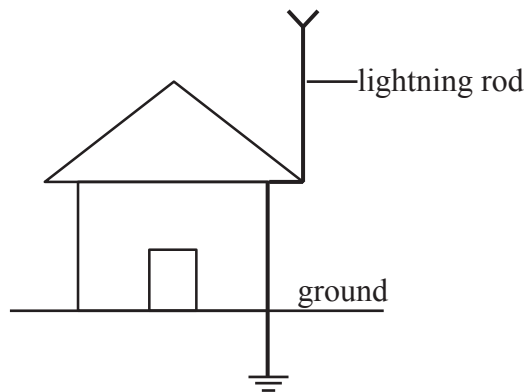


Fig. 11.1

- (i) Explain how the rod protects the building during lightning strikes. [2]
 - (ii) Compare the scientific method with traditional beliefs about lightning protection. [2]
 - (iii) Suggest how combining both approaches in (ii) improves safety in rural communities. [2]
- (b) **Fig. 11.2** shows a circuit diagram consisting of a fuse connected in series with a bulb and a switch.

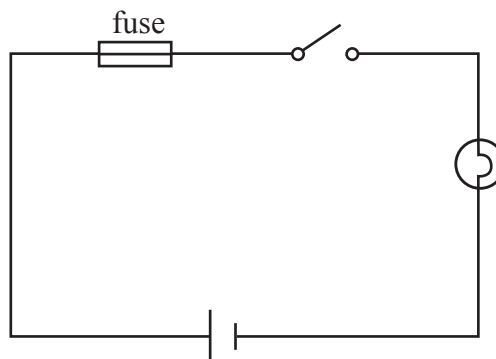


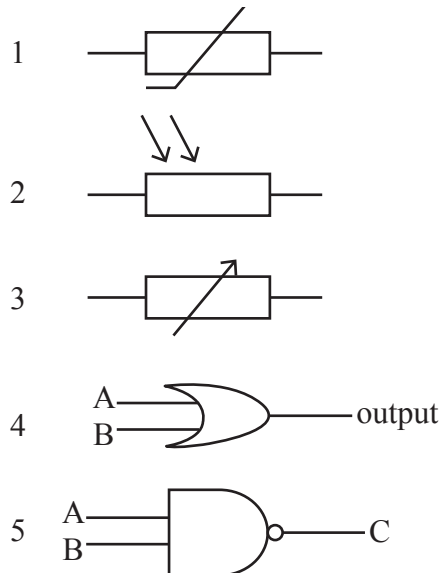
Fig. 11.2

- (i) State the function of the fuse. [2]
- (ii) Explain why fuses are important in protecting homes from electrical fires. [2]

- (c) During a science exhibition competition, a student constructed a simple generator by using a coil, rotating bicycle wheel and attached magnets.

- (i) State the energy conversion that occurs in a generator. [1]
- (ii) State the other name for a generator. [1]
- (iii) Illustrate how the student produced the generator. [3]
- (iv) List **three** ways of increasing the efficiency of the generator. [3]
- (v) Explain how this construction of the generator promotes innovation for science students. [2]

- 12 (a) (i) Identify the symbols:



[5]

- (ii) State which logic gate is known as an inverter. [1]

- (iii) **Fig. 12.2** shows a system at a school that allows lights to turn on manually using a switch at any time or automatically using a light sensor.

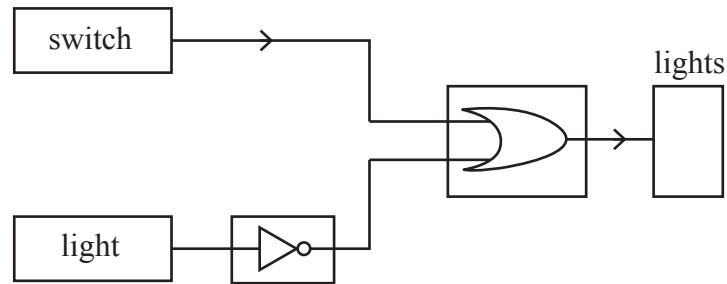


Fig. 12.2

Draw the truth table for the system.

[4]

- 12 (b) A student designed a robot (*vegebot*) to harvest cabbages. The designed robot is shown in **Fig. 12.3**.

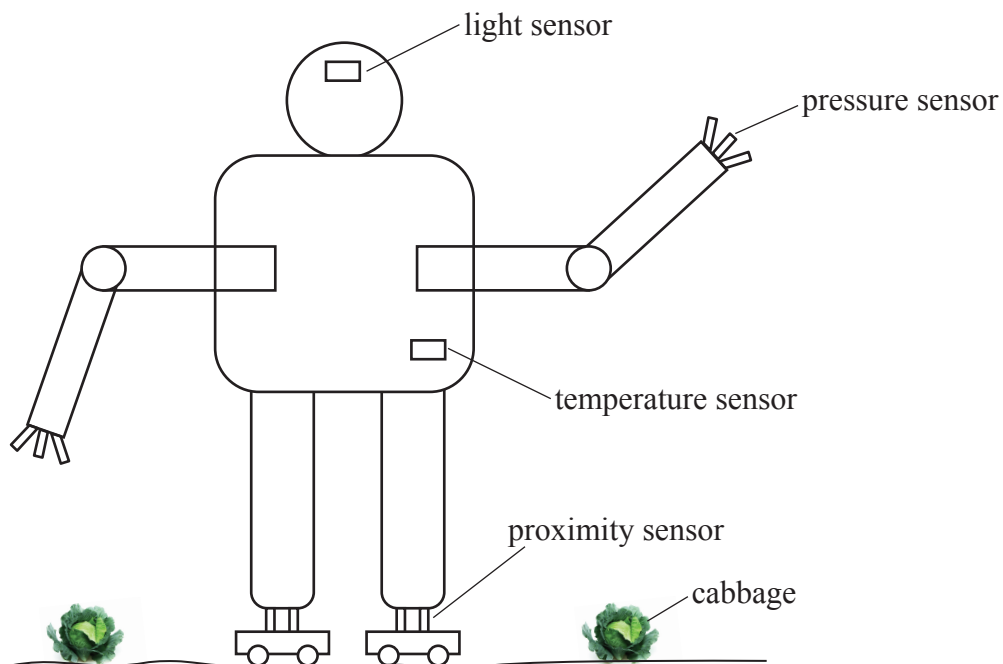


Fig. 12.3

- (i) Describe and explain how each sensor is important for the robot to carry out a harvest. [7]
- (ii) State any **three** advantages of using the robot in harvesting. [3]